

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0429

Pendleton 100 kV
December 26, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0429. The Interconnection Customer proposes a 113.6 MW solar photovoltaic facility interconnecting at the Pendleton 100 kV substation in the Indiana area. The request was received on February 01, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 187 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$420,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Hansel - Butler Municipal Power Plant 100.0 kV
Contingency: E Methuen 500 kV/230 kV XFMR
Loading: 105.2% (Rating: 128 MVA)
DFAX: 5.3%
Pre-project loading: 100.5%
- Facility: Thousand Springs - Hansel Substation 100.0 kV Line
Contingency: East Methuen 500/230 kV XFMR
Loading: 101.4% (Rating: 142 MVA)
DFAX: 5.3%
Pre-project loading: 97.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hansel - Butler Municipal Power Plant 100.0 kV	Reconductor line	\$2,076,000	11% (\$236,000)
Thousand Springs - Hansel Substation 100.0 kV Line	Reconductor line	\$1,620,000	11% (\$184,000)
	Total allocated to project		\$420,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.